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BIO201 Cell Biology

FINAL TERM

CURRENT PAPERS

SPRING 2021

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Oncogenes :

An oncogene is a gene that has the potential to cause cancer. In tumor cells, these genes are often mutated, or expressed at high levels.

LDL Particles : (3)

Low-density lipoprotein particles (LDL-P) are bi-products of fat transport that remain in circulation for an extended time. While in circulation, LDL-Particles can penetrate the artery wall and get stuck, forming a fatty plaque. These plaques can build over time and lead to blockages, resulting in heart attacks and strokes.

- **Normal LDL size** (> 25.5 nm, Normal LDL group)
- **Small LDL** (</= 25.5 nm, Small LDL group)

Housekeeping Cells : (3)

In cell biology, housekeeping genes are typically constitutive genes that are required for the maintenance of basic cellular function and are expressed in all cells of an organism under normal and patho-physiological conditions.

Examples :

- Aactin
- GAPDH
- Ubiquitin

Functions of T cells : (3)

- T cells are responsible for cell-mediated immunity
- T cells directly killed infected host cells
- T cells are the helper cells that activating other immune cells
- T cells producing cytokines
- T cells regulating the immune response

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- T cells are the helper cells that defence against both extracellular and intracellular pathogens.

Epigenetics : (2)

The study of heritable changes in gene function that do not involve changes in DNA sequence but they can change how body reads a DNA sequence. Epigenetic changes alter the physical structure of DNA.

Example :

DNA methylation

Phagocytosis : (2)

Phagocytosis is a process by which particles, microbes or fragments of dead cells are engulfed and internalized, usually by specific membrane receptors.

OR

Phagocytosis is a process by which certain living cells called phagocytes ingest or engulf other cells or particles. The phagocyte may be a free-living one-celled organism such as an amoeba or one of the body cells such as a white blood cell.

Cytokinesis : (2)

Cytokinesis is a physical process of cell division, that normally takes place after mitosis. Cytokinesis is the physical division of the cell cytoplasm, the cell membrane, and cell

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organelles in eukaryotic cells to produce two distinct cells at the end of the cell cycle in both mitosis and meiosis.

Roles of Mitochondria : (3)

Mitochondria is the "Powerhouse" of the cell. They play a vital role in converting food energy into energy for biological processes. Mitochondria also play a vital role in activities that enable cells to function and helps to maintain a healthy body.

- Production of ATP
- Calcium Homeostasis
- Regulation of Innate Immunity
- Programmed Cell Death
- Stem Cell Regulation

Operon : (2)

An operon is a functioning unit of DNA containing a cluster of genes under the control of a single promoter. Operons are of two types, inducible and repressible.

Difference between simple and gram staining ? (2)

Simple Staining :

Simple staining uses and impart only one colour to all bacterial cells. It reveals the size ,

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shape and arrangement of bacterial cells. **E.g** : Methylene Blue

Gram Staining :

Gram staining is a common technique used to differentiate two large groups of bacteria based on their different cell wall components. The Gram stain procedure distinguishes between Gram positive and Gram negative groups by coloring these cells red or violet.

Centrifuge machine and its function :

A centrifuge is a machine that uses centrifugal force to separate various components of a fluid based on their density. This is achieved by spinning the fluid at high speed within a container, thereby separating fluids of different densities or liquids from solids.

Antigens & Antibodies :

Antigen :

An **antigen** is any substance that causes your immune system to produce antibodies against it. Antigens include toxins, chemicals, bacteria, viruses, or other substances that come from outside the body.

Antibody :

An **antibody** is a protein produced by the body's immune system in response to the presence of a foreign substance, called antigens. 5 classes of antibodies IgA , IgD , IgE , IgG or IgM.

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Immunological Memory :

Immunological memory is the ability of the immune system to respond more rapidly and effectively to pathogens that have been encountered previously, and reflects the pre-existence of a clonally expanded population of antigen-specific lymphocytes.

Complement System :

The complement system also known as complement cascade. It is a part of the immune system that enhances the ability of antibodies and phagocytic cells to clear microbes and damaged cells from an organism, promote inflammation, and attack the pathogen's cell membrane.

Q. Which stage of mitosis is the reverse of prophase ?

The last stage of mitosis is telophase which is the reverse of prophase. When the two sets of halved chromosomes have reached their destination, the spindle disappears and the nuclear membrane is formed around each new nucleus.

Q. How PKB promote cell survival ?

Protein kinase B, also known as Akt, is the collective name of a set of three serine/threonine-specific protein kinases that play key roles in multiple cellular processes such as glucose metabolism, apoptosis, cell proliferation, transcription, and cell migration.

OR

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Pkb activated by phosphorylation on amino acids and they are promote the cell by phosphonisitide3 kinase and it is in activation of pro apoptotic factors.

Q. Define epienergetics ?

Epienergetics is the conscious and ecological use of energy to produce the extraordinary.

Q. What is bioartificial tissue?

Bioartificial tissues are made by seeding stem cells or differentiated cells into a natural or artificial biomaterial scaffold shaped in the appropriate form and then by implanting the construct in place of the damaged tissue or organ.

Kin Recognition :

Kin recognition, also called kin detection, is an organism's ability to distinguish between close genetic kin and non-kin. In evolutionary biology and psychology, such an ability is presumed to have evolved for inbreeding avoidance.

Q. Difference between homochromatin, heterochromatin and euchromatin?

Homochromatin :

Regions of a chromosome that are bundled in highly condensed chromatin are known as homochromatin.

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Heterochromatin :

Heterochromatin is such part of the chromosomes, which is a firmly packed form of chromatin and are genetically inactive.

Euchromatin :

Euchromatin is an uncoiled and loosely packed form of chromatin and are genetically active.

Q. Function of incubator :

An incubator is designed to provide a safe, controlled space for infants to live while their vital organs develop. Unlike a simple bassinet, an incubator provides an environment that can be adjusted to provide the ideal temperature as well as the perfect amount of oxygen, humidity, and light.

Q. Function of endoplasmic reticulum ?

The endoplasmic reticulum $\left(\mathrm{ER}\right)$ is the largest membrane-bound organelle in eukaryotic cells and performs a variety of essential cellular functions, including protein synthesis and processing, lipid synthesis, and calcium $\left(\mathrm{Ca2+}\right)$ storage and release.

Q. What is the role of EGF ?

Epidermal growth factor $\left(\mathrm{EGF}\right)$ is a common mitogenic factor that stimulates the proliferation of different types of cells, especially fibroblasts and epithelial cells. EGF

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enhances the differentiation, maturation and survival of a variety of neurons.

Q. N-terminal ER signal sequence?

Signal sequences are N-terminal extensions of newly synthesized secretory and membrane proteins. In eukaryotes, signal sequences direct the insertion of proteins into the membrane of the endoplasmic reticulum and are usually cleaved off by signal peptidase.

Q. What are cargo proteins and functions ?

Any protein that is carried within the vesicles of a cell's secretory system (i.e. from the endoplasmic reticulum to the Golgi apparatus to the plasma membrane).

Function :

Cargo proteins facilitate the formation of transport vesicles in the cytoplasm to vacuole targeting pathways.

Q. Name of antibody that appear on the surface of developing B cell? (3)

IgM is not only the first class of antibody to appear on the surface of a developing B cell. It is also the major class secreted into the blood in the early stages of a primary antibody response, on first exposure to an antigen.

Q. Define Smooth endoplasmic reticulum ?

The smooth endoplasmic reticulum, or smooth ER, is an organelle found in both animal cells and plant cells.The smooth ER also regulates and releases calcium ions and processes

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toxins. It is described as 'smooth' to distinguish it from rough ER, which has ribosomes for protein synthesis on its surface.

Q. Function of ERGIC53 protein ?

The protein is a mannose-specific lectin and is a member of a novel family of plant lectin homologs in the secretory pathway of animal cells.ERGIC-53 and MCFD2 form a protein complex and serve as a cargo receptor to transport FV and FVIII from the ER to the ERGIC and then the Golgi.

Q. Write down the structure of antibody ?

Antibodies are immune system-related proteins called immunoglobulins. Each antibody consists of four polypeptides– two heavy chains and two light chains joined to form a "Y" shaped molecule.This variable region, composed of 110-130 amino acids, give the antibody its specificity for binding antigen.

Q. What are transducer proteins ?

The transducer proteins are proteins that receive and help process information so that proper action by the cell ensues.Two particular types of transducers are described, the sensor kinase of the two-component signal transduction system and the methylated receptors of chemotaxis in bacteria.

Q. Difference between Cytotoxic T Cells and Helper T Cells ?

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Cytotoxic T cells refer to a type of immune cell that can kill certain cells, including foreign cells, cancer cells, and cells infected with a virus. Whereas, the helper T cells refer to a type of T cell that provides help to other cells in the immune response by recognizing foreign antigens and secreting cytokines that activate T and B cells.

Q. Enlist strategies of transcription regulations ?

First, transcription is controlled by limiting the amount of mRNA that is produced from a particular gene. The second level of control is through post-transcriptional events that regulate the translation of mRNA into proteins. Even after a protein is made, post-translational modifications can affect its activity.

Q. Define Cell culture ?

Cell culture is the process by which cells are grown under controlled conditions, generally outside their natural environment. After the cells of interest have been isolated from living tissue, they can subsequently be maintained under carefully controlled conditions.

OR

Cell culture refers to the removal of cells from an animal or plant and their subssequent growth in a favourable artificial environment.

Q. Define lectin pathway ?

The lectin pathway or lectin complement pathway is a type of cascade reaction in the

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complement system, similar in structure to the classical complement pathway, in that, after activation, it proceeds through the action of C4 and C2 to produce activated complement proteins further down the cascade.

Q. Define Paracrine signaling ?

Paracrine signaling is a form of cell signaling, a type of cellular communication in which a cell produces a signal to induce changes in nearby cells, altering the behaviour of those cells.Cells that produce paracrine factors secrete them into the immediate extracellular environment.

Q. Which mechanism is used by the birds for migration?

Birds use a variety of different compass mechanisms for orientation during migration, based on celestial or geomagnetic cues. Celestial compass cues (stars, sun, skylight polarization patterns) provide birds with directional information relative to a true geographic reference. E.g., geographic North [7,8,9,10,11,12].

Q. Difference between multiple transmembrane protein and double transmembrane protein ?

1. Multiple transmembrane protein have multiple topogenic sequence and double transmembrane absent is multiple topogenic sequence.
2. In multiple transmembrane protein alpha helix each membrane rather than double transmembrane protein helix and beta barrels.

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Q. Which element promote the autonomous cell growth in cancer cells?

Autonomous cell growth for cancer cells are polypeptides that stimulates the cell poliferation.

Q. What are the advantages of storing cells at low temperature ?

1. The structure of cell is not degerative.
2. It reduces the risk of microbial contamination.
3. It reduces cross contamination with other cell samples or tissues.
4. At low temperature cells can be preserved for long time.
5. It help cells to prevent dehydration.

Q. What are proto oncogene ?And write its classes ?

A gene involved in normal cell growth is called proto oncogene. Proto-oncogenes encode proteins that function to stimulate cell division and cell growth. The mutations in a proto-oncogene may cause it to become an oncogene, which can cause the growth of cancer cells.

OR

A proto oncogene is a normal gene that can become an oncogene , either after mutation or increased expression. Proto oncogenes codes for proteins that help to regulate cell growth

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and differentiation.

Classes :

- Growth promoting oncogenes
- Growth inhibiting cancer suppressor genes
- DNA repairing genes
- Apoptosis genes

Q. Role of EGF-153 protein ?

Epidermal growth factor is a 153 amino acid cytokine which is cleaved from a very large integral membrane protein precursor. EGF-153 acts to stimulate growth of epithelial cells, accelerate tooth cruption , eyelid opening in mice , inhibit gastric acid secretion and is involved in wound healing.

Q. How DNA packed in cells ?

DNA is tightly packed up to fit in the nucleus of every cell. A DNA molecule wraps around histone proteins to form tight loops called nucleosomes. These nucleosomes coil and stack together to form fibers called chromatin.

E.coli Signalling :

In the LuxR process , E.coli detect and autoinducer synthesized by other types of bacteria

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but during the LuxS signalling system , E.coli recognized self produced autoinducer signals produced by other microbes or the human stress hormone (epinephrine).

Q. Differentiate between Homotypic and Heterotypic fusion ?

- Homotypic fusion occurs when the two membranes merging together are of the same type i.e : Vacuole to vacuole fusion.
- Heterotypic fusion when the two membranes merging together are of the different type i.e : When a andosomal membrane fuse with golgi membrane.

Q. How UV light help in sterilization process ?

UV sterilization also known as UV disinfection or ultraviolet germicidal irradiation (UVGI) works by breaking down certain chemical bonds and scrambling the structure of DNA, RNA and proteins, causing a microorganism to be unable to multiply by damaging the DNA of their cells.

Q. What is laminar flow cabinet ? Also write its applications?

A laminar flow cabinet or tissue culture hood is a carefully enclosed bench designed to prevent contamination of semiconductor wafers, biological samples, or any particle sensitive materials. Air is drawn through a HEPA filter and air is passed through a HEPA (High Efficiency Particulates Air) filter which removes all airborne contamination to maintain sterile conditions. The function of laminar flow cabinet to protect the product or process from particles that are trapped in the HEPA filter.

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Applications of laminar flow cabinet :

Laminar flow cabinets are used in laboratories for contamination sensitive processes like plant tissue culture. Other laboratories processes like media plate preparation and culture of organisms can be performed inside the cabinet. Operations of particle sensitive electronic devices are performed inside the cabinet.

Q. What is Adaptive immune system and write its function ?

The adaptive immune system, also referred as the acquired immune system, is a subsystem of the immune system that is composed of specialized, systemic cells and processes that eliminate pathogens or prevent their growth. Adaptive immunity is consists of lymphocytes, either T or B cells, and includes both CD8+ cytotoxic T cells.

Function adaptive immune system :

The function of adaptive immune responses is to destroy invading pathogens and any toxic molecules they produce. Because these responses are destructive, it is crucial that they be made only in response to molecules that are foreign to the host and not to the molecules of the host itself.

Positive & Negative control of transcription :

- When transcription is under **positive control**, a protein known as an activator binds to the DNA in order for transcription to take place

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- When transcription is under **negative control**, a protein known as a repressor binds to the DNA and blocks transcription.

Q. What is synapse and its steps involved in the mechanism?

A synapse is a small gap at the end of a neuron that allows a signal to pass from one neuron to the next. Synapses are found where nerve cells connect with other nerve cells.

Steps :

- Synthesis and storage of neurotransmitters
- Neurotransmitter release
- Neurotransmitter postsynaptic receptors
- Inactivation of Neurotransmitters

Q. Difference between confluent monolayer and past confluence layer ?

Confluent monolayer refers to cells in tissue culture, e.g. epithelial cells, which form a cohesive sheet comprising of a single cell layer filling the entire surface area of the bottom of the culture dish. Cells have then reached confluence.

Q. What is the process of t cell activation ?

The T cell encounters a dendritic cell $\left(\mathrm{DC}\right)$ bearing its cognate peptide in an MHC molecule, and binds the peptide-MHC though CD3 and CD4 or 8. Subsequently,

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co-stimulation occurs through DC-bound CD86, CD80, OX40L and 4-1BBL. This induces full activation and effector function in the T cell.

Q. How caM kinase II activity helps in muscular movement?

caM kinase II is a serine specific protein kinase that is regulated by the

Ca2+/calmodulin $\left(Ca2+/CaM\right)$. caM kinase II is involved in many signalling cascades and are important mediators of signal transduction events triggered by stimuli that increase intracellular levels of free $\left[Ca2+\right]$.

Q. Write down the types of mutations ? (5)

Mutations:

A Mutation occurs when a DNA gene is damaged or changed in such a way as to alter the genetic information carried by that gene.

Somatic mutations :

Occur in somatic $\left(body\right)$ cells & passed on to the daughter cells after mitosis but not passed on to the children.

Germ line mutations:

Occur the germ line cells that give rise to gametes. A gamete with the mutation passes it on to embryo after fertilization.

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Point mutations:

Mutations of single base pairs.

Chromosomal mutations:

Change the position or orientation of a DNA segment.

Silent Mutations:

Mutations result in no change in amino acids due to redundancy genetic code.

Missense Mutations:

Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein.

Frame-Shift Mutations:

No substitutions. Single base inserted or deleted from DNA. Genetic message out of register "reading frame" changed.

Q. Difference between dna and cdna ?

Definition :

DNA: DNA refers to a type of nucleic acid, consisting of a double-helix, which is held by the hydrogen bonds between the purines and pyrimidines in the two chains.

cDNA: cDNA refers to the DNA that is synthesized using messenger RNA as the template.

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Names :

DNA: DNA refers to deoxyribonucleic acid.

cDNA: cDNA refers to complementary DNA.

Synthesized from :

DNA: DNA is synthesized from the existing genomes.

cDNA: cDNA is synthesized from the cytosolic mRNA.

Synthesized by :

DNA: DNA is synthesized during DNA replication by the action of the enzyme DNA polymerase.

cDNA: cDNA is synthesized during reverse transcription by the action of the enzyme reverse transcriptase.

Single/Double-stranded :

DNA: DNA is double-stranded in nature.

cDNA: cDNA is single-stranded.

Consist of :

DNA: DNA consists of both coding and non-coding sequences of an organism.

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cDNA: cDNA only consists of the coding regions or the exons.

Called as :

DNA: The total DNA of an organism is called the genome.

cDNA: The total cDNA of an organism is called the transcriptome.

Quantity :

DNA: DNA contains a large number of base pairs.

cDNA: cDNA contains a few base pair compared to DNA.

Use :

DNA: DNA can be used to produce genomic libraries.

cDNA: cDNA can be used to produce cDNA libraries.

Q. How G protein is involved in formation of GDP ?

G proteins are important signal transducing molecules in cells. Receptors stimulate GTP binding (turning the G protein on). RGS proteins stimulate GTP hydrolysis (creating GDP, thus turning the G protein off).

Q . What are Cell line ?

A cell line is a permanently established cell culture that will proliferate indefinitely given appropriate fresh medium and space.

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Q. Define Macrophages ?

Macrophage is a type of white blood cell that surrounds and kills microorganisms, removes dead cells, and stimulates the action of other immune system cells. They can also present antigens to T cells and initiate inflammation by releasing molecules known as cytokines that activate other cells.

Q. Define lysosomes ? And its function .

A lysosome is a membrane-bound cell organelle that contains digestive enzymes. Lysosomes are involved with various cell processes. They break down excess or worn-out cell parts. If the cell is damaged beyond repair, lysosomes can help it to self-destruct in a process called programmed cell death, or apoptosis.

Function :

Lysosomes function as the digestive system of the cell, serving both to degrade material taken up from outside the cell and to digest obsolete components of the cell itself.

Q. Write reagents used in the sterilization or equipments? Names atleast 5 .

Chemical reagents used in sterilization :

- Ethylene Oxide.
- Ozone.
- Bleach.

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- Glutaraldehyde and Formaldehyde.
- Phthalaldehyde.
- Hydrogen Peroxide.
- Peracetic Acid.
- Silver.

Equipments used in sterilization :

- Steam sterilizers.
- Sterlization system.
- Control system.
- Composite trays.
- Gas-plasma sterilization.
- Aseptic environments.
- Steam generators.
- E-beam system.

Q. Define Metaphase and Telophase ?

Metaphase :

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Metaphase is the third phase of mitosis, the process that separates duplicated genetic material carried in the nucleus of a parent cell into two identical daughter cells. Long protein filaments called kinetochore microtubules extended from poles on either end of the cell and attached to the kinetochores.

Telophase :

Telophase is the fifth and final phase of mitosis, the process that separates the duplicated genetic material carried in the nucleus of a parent cell into two identical daughter cells. Telophase begins once the replicated, paired chromosomes have been separated and pulled to opposite sides, or poles, of the cell.

Q. What is the role ok P13K ?

mTOR is well-known as an indirect PI3K effector involved in mitogenesis. It plays an essential role for numerous cell functions, such as proliferation and cell growth, and its deregulation can lead to tumor growth, angiogenesis, and metastasis

Q. Define Adult stem cells and their beneficial and theraputic uses?

Adult stem cells are undifferentiated cells, found throughout the body after development, that multiply by cell division to replenish dying cells and regenerate damaged tissues. Also known as somatic stem cells, they can be found in juvenile, adult animals, and humans, unlike embryonic stem cells. Adult stem cells are involved in regenerating tissues for renewal and damage repair. Benefits of adult stem cells is that they are important for

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maintainance and repair of damage organ and tissue and they restore dying cells. The therapeutic use of adult stem cell is that stem cell transplant cells that are damaged by chemotherapy.

Q. Define Haemocytometer ?

The hemocytometer or haemocytometer is a counting-chamber device originally designed and usually used for counting blood cells. The hemocytometer was invented by Louis-Charles Malassez and consists of a thick glass microscope slide with a rectangular indentation that creates a precision volume chamber.

Q. What are cAmP ?

Cyclic adenosine monophosphate (cAMP, cyclic AMP, or 3',5'-cyclic adenosine monophosphate) is a second messenger important in many biological processes. cAMP is a derivative of adenosine triphosphate (ATP) and used for intracellular signal transduction in many different organisms, conveying the cAMP-dependent pathway.

Q. Which cells are responsible for cell mediated response ?

T cells are responsible for cell-mediated immunity, which involves the destruction of body cells that are infected with pathogens or have become damaged or cancerous. T cells are activated when they encounter an antigen-presenting cell matching their own receptor.

Q. Write a note on stages of cell cycle ?

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A cell cycle is a series of events that takes place in a cell as it grows and divides.Cell cycle has different stages called G1, S, G2, and M. G1 is the stage where the cell is preparing to divide. After this, it then moves into the S phase where the cell copies all the DNA.

G1 phase:

G1 phase also, first gap which is a first phase of interphase centered on cell growth during mitosis.

S phase :

S phase second or synthesis stage of interphase during which DNA replication occurs.

G2 phase:

G2 phase is also, second gap or third phase of interphase during which the cell undergoes final preparations for mitosis.

Interphase :

Interphase is a period of the cell cycle leading up to mitosis; includes G1, S, and G2 phases the interim period between two consecutive cell divisions.

M phase :

M phase consists of nuclear division (karyokinesis). It is a relatively short period of the cell cycle. M phase is complex and highly regulated.

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karyokinesis:

It is a mitotic nuclear division.

Mitosis:

Mitosis also called karyokinesis which is a period of the cell cycle during which the duplicated chromosomes are separated into identical nuclei; includes prophase, prometaphase, metaphase, anaphase, and telophase.

Cytokinesis:

Cytokinesis division of the cytoplasm following mitosis that forms two daughter cells.

Q. What are the clinical uses of cell culture ?

Cell culture includes processes for isolating cells from tissue and maintaining, proliferating, and/or differentiating the cells in a culture vessel containing medium.Cultured cells are not only used for basic research in biology and molecular biology, but also for drug discovery and regenerative medicine.

Q. What is palindromic sequence ?

A palindromic sequence is a nucleic acid sequence in a double-stranded DNA or RNA molecule wherein reading in a certain direction $\left(\text{e.g. } 5' \text{ to } 3'\right)$ on one strand matches the sequence reading in the opposite direction $\left(\text{e.g. } 3' \text{ to } 5'\right)$ on the complementary strand.

Q. Name three RNA viruses ?

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Human diseases causing RNA viruses include Orthomyxoviruses, Hepatitis C Virus (HCV), Ebola disease, SARS, influenza, polio measles and retrovirus including adult Human T-cell lymphotropic virus type 1 (HTLV-1) and human immunodeficiency virus (HIV).

Q. Define Active and Passive Transport ? (3)

Active Transport is the movement of molecules or ions from an area of lower to higher concentration.

Passive Transport is the movement of molecules or ions from an area of higher to lower concentration.

Q. What is primary and secondary cell culture? (2)

Cells for primary cell culture are obtained directly from an animal or plant tissue while cells for secondary cell culture are obtained from an already established primary culture.

Q. What is primary cell culture ? (3)

Primary cell culture is the ex vivo culture of cells freshly obtained from a multicellular organism, as opposed to the culture of immortalized cell lines.When whole or partial tissues are isolated and maintained ex vivo, the procedure is termed primary tissue culture.

Q. Define Mutagens ?

A Mutagen is an agent of substance that can bring about a permanent alteration to the

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physical composition of a DNA gene such that the genetic message is changed. **E.g** : Chemical Mutagens ,Radiation ,Sunlight.

Q. Difference between B cells and T cells ?

B cells can connect to the antigen on the surface of the invading virus or bacteria. The B cells are made in the bone marrow and also mature there. B cells are responsible for antibody-mediated immunity.

T cells which can only connect to virus antigens on the outside of infected cells. The T cells are made in the bone marrow and travel to the thymus and mature there.T cells are responsible for cell-mediated immunity.

Q. Difference between purine and pyrimidine ?

Purines and pyrimidines are the nitrogen bases that hold DNA strands together through hydrogen bonds.The purines in DNA are adenine and guanine, the same as in RNA. The pyrimidines in DNA are cytosine and thymine; in RNA, they are cytosine and uracil. Purines are larger than pyrimidines because they have a two-ring structure while pyrimidines only have a single ring.

Q. Write functions of growth promoting oncogenes ?

Oncogenes are a structurally and functionally heterogeneous group of genes, whose protein products act pleiotropically and affect multiple complex regulatory cascades within the cell. They regulate cell proliferation, growth, and differentiation, as well as control of the

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cell cycle and apoptosis.

Q. What is the Difference Between Stem Cells and Specialised Cells ?

Stem cells refer to the undifferentiated cells of a multicellular organism capable of giving rise to indefinitely more cells of the same type and from which certain other kinds of cell arise by differentiation while specialised cells refer to the cells in multicellular organisms modified to carry out a particular function, such as transporting a certain substance or executing a specific task.

Q. Difference between MCH-I and MHC-II ?

MHC-I :

1. Have 8-10 amino acids
2. Peptide binding domain alpha-1, alpha-2
3. Present antigen to CD 8 T- cells
4. Found on all surfaces of nucleated cells
5. Composed of one peptide encoded in HLA locus and beta 2 microglobullin
6. Bind with endogenous antigen
7. Has no invariant chain

MHC-II :

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1. Have 13-18 amino acids
2. Peptide binding domain alpha1, beta1
3. Present antigen to CD 4 T- cells
4. Found on surface of APCs and activated T cells
5. Composed of two peptide encoded by HLA locus
6. Bind with exogenous antigen
7. Has invariant chain

Q. ER signal sequence is guided to the ER by which of the two components ?

1. SRP which cycles between the ER membrane and the cytosol and binds to the signal sequence.

2. SRP receptor in the ER membrane.

Q. Name 5 major growth promoting oncogenes ?

1. Growth factors

2. Growth factor receptors

3. Signal transducing proteins

4. Nuclear transcription factors

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5. Cyclins dependent kinases CDKs

Q. Difference between aneuploidy and polyploidy ?

Aneuploidy refers to a numerical change in part of the chromosome set.

Polyploidy refers to a numerical change in the whole set of chromosomes.

Q. What is the function of rod cells ?

Rod cells are a type of photoreceptor cells found in the retina of eyes. These cells are capable of absorbing light and converting the light into signals that can trigger a change in the membrane potential which results in visual phototransduction.

Q. Write down the 3 steps of RNA processing ?

1. Addition of 5' cap

2. Addition of 3' poly-adenylation tail

3. Intron splicing

Q. Write down the mechanism of growth promoting oncogenes?

The activation of oncogenes involves genetic changes to cellular proto-oncogenes. The consequence of these genetic alterations is to confer a growth advantage to the cell. Three genetic mechanisms activate oncogenes in human neoplasms: mutation, gene amplification and chromosome rearrangements.

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Q. How beta blockers effects the function of adrenaline?

Beta blockers work by blocking the effects of the hormone epinephrine, also known as adrenaline. Beta blockers cause the heart to beat more slowly and with less force, which lowers blood pressure. Beta blockers also help widen veins and arteries to improve blood flow.

Q. What are TGF-β receptors family ?

The transforming growth factor beta $\left(\text{TGF}\beta\right)$ receptors are a family of serine/threonine kinase receptors involved in TGF beta signaling pathway. These receptors bind growth factor and cytokine signaling proteins in the TGF-beta family such as TGFβs $\left(\text{TGF}\beta1, \text{TGF}\beta2, \text{TGF}\beta3\right)$. TGFβ family receptors are grouped into three types, type I, type II, and type III.

Q. Define cloning ?

Cloning is the process of generating a genetically identical copy of a cell or an organism.

Q. What is meant by polarity of water molecules ?

Water is a "polar" molecule, meaning that there is an uneven distribution of electron density. Water has a partial negative charge near the oxygen atom due the unshared pairs of electrons, and partial positive charges near the hydrogen atoms.

Q. Why the cell is important ?

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Cells are the basic building blocks of all living things. The human body is composed of trillions of cells. They provide structure for the body, take in nutrients from food, convert those nutrients into energy, and carry out specialized functions.

Q. What is the cytoskeleton ?

Cytoskeleton, a system of filaments or fibres that is present in the cytoplasm of eukaryotic cells. Three major types of filaments make up the cytoskeleton: actin filaments, microtubules, and intermediate filaments.

Q. Write down the types of growth promoting oncogenes ?

Growth Factors :

Some cells with oncogenes become self-sufficient by synthesizing the growth factors to which they respond.

Growth Factor Receptors :

Oncogenes may activate or increase growth factor receptors on the surface of cells to which growth factors bind.

Signal Transduction Proteins :

Other oncogenes affect proteins involved in transmitting signals from the receptor of the cell to the nucleus.

Transcription Factors :

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Transcription factors are proteins that regulate when cells enter, and how they progress through the cell cycle.

Cyclins dependent kinases CDKs :

Cell cycle control proteins are products of oncogenes that can affect the cell cycle in a number of different ways. Some, such as cyclin D1 and cyclin E1 work to progress through specific stages of the cell cycle,

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